

Government Engineering College, Thrissur Department of Electronics and Communication Engineering					
PROPOSAL FOR SEMINAR					
Course Code	ECQ413	Semester	VII	Program	BTECH - ECE
Course Name	SEMINAR	Batch	2022 - 2026	Proposal No.	
Reg. No.	TCR22EC066	Name of the student		VISHNU DIVAKAR	
Title of Seminar	Recent Advances in MIMO Antenna Design for 5G and Radar Systems				
Relevance of the topic	MIMO antenna innovations enable fast, low-latency 5G and radar systems—vital for autonomous vehicles, IoT, and smart infrastructure.	Technical Contents		Reconfigurable Arrays, DGS Isolation, ML-Optimized DRAs, Metamaterials, Dual-Band mmWave, AI Beamforming, Real-Time 5G/Radar Use	
Name of the guide	Dr. JOB CHUNKATH			Accept	Reject
Give an Abstract of the Seminar limited to 200 – 300 words					
Abstract					
- The emergence of 5G and intelligent radar systems has accelerated innovation in Multiple-Input Multiple-Output (MIMO) antenna design. MIMO technologies offer high throughput, spatial diversity, and interference mitigation, which are essential for modern wireless applications. This seminar presents an overview of recent advancements in MIMO antenna structures aimed at supporting ultra-reliable, low-latency communication in 5G and radar platforms. - A key highlight includes the development of reconfigurable MIMO antennas operating at 24 GHz, which utilize split-ring resonator metamaterials and varactor diodes to enable dynamic frequency tuning and reduce mutual coupling. These designs achieve exceptional isolation (>38 dB) and low envelope correlation coefficients ($ECC < 0.0004$), making them ideal for radar and autonomous vehicular systems. - Complementing these innovations are compact dual-band MIMO designs with defected ground structures (DGS) and tapered feeds, enabling efficient operation in 3.4–5.1 GHz and mmWave bands. Furthermore, machine learning algorithms are increasingly employed to optimize antenna dimensions, placement, and performance in real-time environments. - Applications of these technologies span 5G base stations, smart surveillance, industrial automation, and cloud robotics, where high gain, miniaturization, and adaptability are essential. For electronics and communication engineers, these advances open up opportunities in RF design, antenna optimization, and intelligent embedded systems.					
References (Give references in IEEE format)					
F. Alwatban, M. M. Ali, and M. T. Islam, "Design and Implementation of Reconfigurable MIMO Antennas for Radar Applications," <i>J. Electr. Eng.</i>, vol. 76, no. 2, pp. 89–97, Jun. 2025. S. Pasha, N. R. Al-Akhras, and J. Hossain, "High-Gain MIMO Beamforming Antenna for DSRC and mmWave 5G Integration," <i>IEEE Access</i>, vol. 13, pp. 50233–50242, Jun. 2025.					

3. M. A. Abdellatif et al., "Compact MIMO Antenna Using C-Shaped Metamaterial for 5G/6G Applications," *Ain Shams Eng. J.*, vol. 25, no. 4, pp. 134–143, Apr. 2025.
4. Z. Zhong, H. Zhou, and R. K. Sharma, "Machine Learning-Aided Design of Dielectric Resonator MIMO Antenna," *Int. J. Microw. Wirel. Technol.*, vol. 15, no. 1, pp. 56–64, Jun. 2025.
5. A. K. Hossain, M. Chowdhury, and T. Roy, "Design of Eight-Element MIMO Antenna Using Defected Ground Structure for 5G Applications," *Int. J. Comput. Inform. Technol.*, vol. 6, no. 1, pp. 45–53, 2025.

Signature of Guide: